Exercise 5: Foundations of Semantic Analysis

Recommended reading: Section 16.3.1 of the Handbook of Model Checking (definition and example of semi-lattices).

1 From Partial Orders to Lattices

a. Which three properties need to be fulfilled by a set $\mathbb{X}$ and subsumption relation $\sqsubseteq$ to make a partial order?

b. Which properties need to be fulfilled by a lattice?

c. Consider these tuples:

1. $(\mathbb{Z}, =)$
2. $(\mathbb{Z}, <)$
3. $([-10, 10], \leq)$ where $[-10, 10]$ is the interval of integers from -10 to 10 (inclusive).
4. $(2^Q, \subseteq)$, Q finite
5. $(\{\text{unknown}, \text{not_null}\}, \sqsubseteq)$ with $x \sqsubseteq y$ iff $x = y$ or $y = \text{unknown}$
6. $(\{\bot, +, -, 0, \top\}, \sqsubseteq)$ with $x \sqsubseteq y$ iff $x = y$ or $x = \bot$ or $y = \top$. The signs denote positive, negative, and zero integer values; $\bot$ denotes no possible value and $\top$ an unknown sign.
7. $(2^X, \sqsubseteq)$, with set X of program variables and $x \sqsubseteq y$ iff $x \cup y \neq \emptyset$.

Decide for each whether it is a partial order.

d. Decide, for each of the tuples $(\mathbb{X}, \sqsubseteq)$ identified in (c) as partial orders, whether it has a finite height. If it does, state that height (potentially with regards to the concrete $\mathbb{X}$).

e. Which of the identified partial orders can be extended into lattices? For those who can, define the $\top$ and $\bot$ element.

2 Upper and Lower Bounds

Notation: $\mathbb{N} = \{0, 1, 2, \ldots\}$.

a. Consider the partial order $(\mathbb{N}, \leq)$. Compute for each of the following subsets the least upper bound and the greatest lower bound if they exist.

1. $\{1, 2\}$

2. $\{2, 4, 8\}$

3. $\emptyset$

4. $\mathbb{N}$

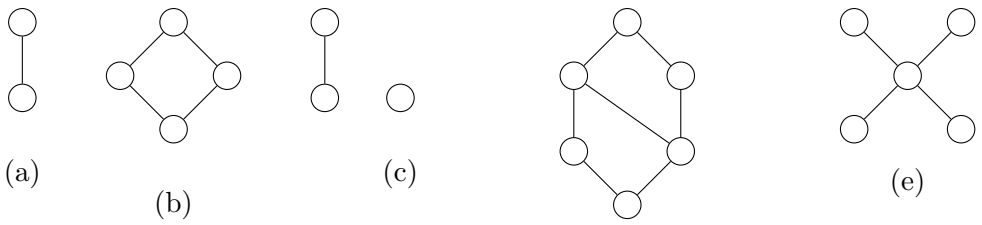
b. Consider the following partial orders. What are the least upper bound and greatest lower bound operators for finite, non-empty subsets?

1. $(2^Q, \subseteq)$

2. $(\mathbb{I}, \subseteq)$ with set $\mathbb{I}$ the set of all intervals $\subseteq 2^{\mathbb{Z}}$.

3 Lattices

Consider the following partial orders. Which of them are lattices?



(a) (b) (c) (d) (e)